54 First-Season Lessons from a FLOSS Pipeline for Trimming Photogrammetric Models

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Photogrammetric 3D models provide an invaluable path toward conserving the inherently deconstructive process of archaeology, capturing excavated features with high precision and exceptional detail. These advantages compound with single-context excavation, where each stratigraphic unit is documented with both topographic and photogrammetric methods. In this way, digital records can be used to reconstruct stratigraphy in three-dimensional space, utilizing an free-to-use 3D engine like Unity. However, the practical use of photogrammetric models remains limited by the labor required to prepare them for deployment, which must be trimmed, decimated, organized, and exported before they can run in interpretive 3D visual environments. To bridge that gap, a FLOSS pipeline was developed to provide a toolkit for trimming and decimating archaeological photogrammetry models, the two key processing steps required for single-context use in a 3D environment. During the 2026 summer field season, the pipeline was employed at the Gabii Project, an ancient urban site outside of Rome and an early adopter of single-context photogrammetric recording. This paper reports how the existing Gabii workflow was modified to utilize this semi-automated pipeline, exploring and documenting how others might do the same. It argues that open-source-based automation necessitates and facilitates the type of standardized good practices that every excavation should strive for, while creating a more efficient and impactful route to the dissemination of archaeological data. The pipeline application and associated scripts are made available in a GitHub repository for anyone attempting to integrate photogrammetric documentation into everyday field recording and interpretation.